ORAL DEFENSE · DEEP LEARNING FINAL PROJECT

⏱ 0:00 – 0:30 (30s)

Finding the Right Law, in Khmer

Can Deep Learning Match Everyday Khmer Questions to Official Statutory Articles?

THE CENTRAL RESEARCH QUESTION

«តើកិច្ចសន្យាត្រូវបង្កើតឡើងដោយរបៀបណា យោងតាមក្រមរដ្ឋប្បវេណី?»

"How is a contract formed under the Civil Code?" — Citizens ask questions in conversational language. Statutes are drafted in formal legal prose across nearly two thousand articles.

Author: Mengchheang Long · Bachelor of Software Engineering, Department of Engineering
Lecturer: Mr. Soklong HIM · Academic Year 2026–2027 · Format: 10-Min Talk + 5-Min Q&A

1,976	4 Models	200 Qs	+44.1%
STATUTORY ARTICLES	SCRATCH TO FINE-TUNED	HUMAN BENCHMARK (T-Q)	MRR GAIN VS BM25 (P < 0.001)

1. MOTIVATION & PROBLEM STATEMENT

⏱ 0:30 – 1:15 (45s)

The Lexical Keyword Trap: Why Keyword Search Fails

Keyword overlap is misleading on formal statutory codes

❌ Concrete Retrieval Failure (BM25)

«តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?»

Target: **Civil Code Art. 336** (Offer and Acceptance) & **Art. 311** (Contract Definition).

BM25 Top Match: Article 623 (*Manner of dividing proceeds in co-ownership*)

Trap: Matched the colloquial question word «របៀប» (manner/how) instead of contract law!

⚠️ Orthographic Challenge: No Word Spaces

Khmer text contains **zero whitespace between words**: «ការបង្កើតកិច្ចសន្យា...». Keyword search relies on CRF tokenizers; a single compound split error breaks term matching.

QUANTITATIVE ERROR AUDIT

57.5%

of BM25's retrieval misses on human questions are caused purely by Vocabulary Mismatch.

✅ The Deep Learning Solution

Project queries and legal articles into a **continuous dense vector space** where proximity reflects semantic meaning rather than shared keywords.

2. DATA ENGINEERING & EXTRACTION

⏱ 1:15 – 2:05 (50s)

Before AI, We Needed Usable Data: The Limon Font Hurdle

Solving 1990s legacy font encoding to extract 1,976 consecutive articles without OCR noise

RAW PDF TEXT LAYER (LIMON ASCII KEYSTROKES)

```
maRta 336›— karbegáItkic©snñaedaysMeNI nig sVIkar
```

→

NORMALIZED LOGICAL UNICODE

មាត្រា ៣៣៦.- ការបង្កើតកិច្ចសន្យាដោយសំណើ និង ស្នើការ

INPUT
Official PDFs
ODC Civil & Criminal

→

STEP 1
Limon Extractor
Byte-level glyph mapping

→

STEP 2
FSM Syllable Engine
Vowel reordering (34 tests)

→

OUTPUT
1,976 Articles
0 missing articles

Audited Article Counts

- **Civil Code 2007:** 1,304 articles (591k characters)
- **Criminal Code 2009:** 672 articles (286k characters)
- **Verification:** 100% consecutive numbering audited with 0 gaps.

Statutory Hierarchy Preserved

គន្ថី (Book) ➔ មាតិកា (Title) ➔ ជំពូក (Chapter) ➔ ផ្នែក (Section) ➔ មាត្រា (Article)

Complete hierarchy retained as metadata for retrieval filtering and context attribution.

3. TRAINING FORMULATION & QUARANTINE

⏱ 2:05 – 2:55 (50s)

Weak Supervision & The Zero-Leakage Protocol

Training without human QA pairs while strictly isolating test evaluation

1. Weak Supervision Training Pairs

Zero public Khmer legal QA datasets exist. We leveraged the inherent structure of statutory articles:

Query: Article Title (*«ការបង្កើតកិច្ចសន្យា...»*)

⬇

Passage: Article Body (with title line stripped via regex)

Removed 270 duplicate titles (e.g. 63 identical penalty titles) to prevent ambiguous labels ➔ 1,471 unique pairs.

2. Zero-Leakage Protocol (Course §5.A)

QUARANTINE SHIELD
All 261 articles cited by the 200 benchmark questions were strictly removed from training and validation!

TRAIN (80%)	VAL (10%)	TEST T-T (10%)
1,176 pairs	147 pairs	148 titles

Deterministic split with fixed seed 42. Test sets evaluated strictly once.

Four Experiments, Four Different Architectural Hypotheses

Testing capacity, pre-training, parameter efficiency, and language specialization

A1: From Scratch

BiLSTM Dual Encoder

1.97M Trainable Params

Word Tokens (CRF)

▼

Embedding (300-d)

▼

1-Layer BiLSTM (512-d)

▼

Masked Mean Pooling

Hypothesis: Can a model learn legal semantics without any pre-training?

A2: Parameter-Efficient

XLM-R Linear Probe

0.59M Trainable (0.21%)

Subword Tokens

▼

Frozen XLM-R (278M)

▼

Mean Pooling

▼

Linear Head (768→768)

Hypothesis: Can a fast 5.9s linear head adapt generic multilingual space?

A3: Winner 🏆

XLM-R Full Fine-Tune

278.04M Trainable Params

Subword Tokens

▼

12-Layer Transformer

▼

All Attention Heads

▼

Mean Pooling + L2

***Hypothesis:* Does full self-attention bridge citizen speech and legal clauses?**

A4: Khmer-Native

PrahokBART Encoder

35.83M Trainable Params

Khmer 32k Vocab

▼

BART Encoder (6 Layers)

▼

Mean Pooling

▼

L2 Normalization

Hypothesis: Does native Khmer pretraining beat larger multilingual models?

Contrastive Metric Learning: InfoNCE in Action

Pulling matching query-article pairs together while repelling in-batch negatives

In-Batch Negative Similarity Matrix (Illustrative)

For query q_i, passage p_i is positive; the remaining $B - 1$ passages in the batch serve as negatives (actual A3 training uses $B = 16 \rightarrow 15$ negatives per query):



Diagonal = Positives · Off-Diagonal = In-Batch Negatives

Symmetric InfoNCE Loss (τ = 0.05)

Maximizes mutual information between query and passage representations:

```
L = - 1/(2B) * [ log(pos / sum_j exp(sim_ij/τ)) + log(pos / sum_j exp(sim_ji/τ)) ]
```

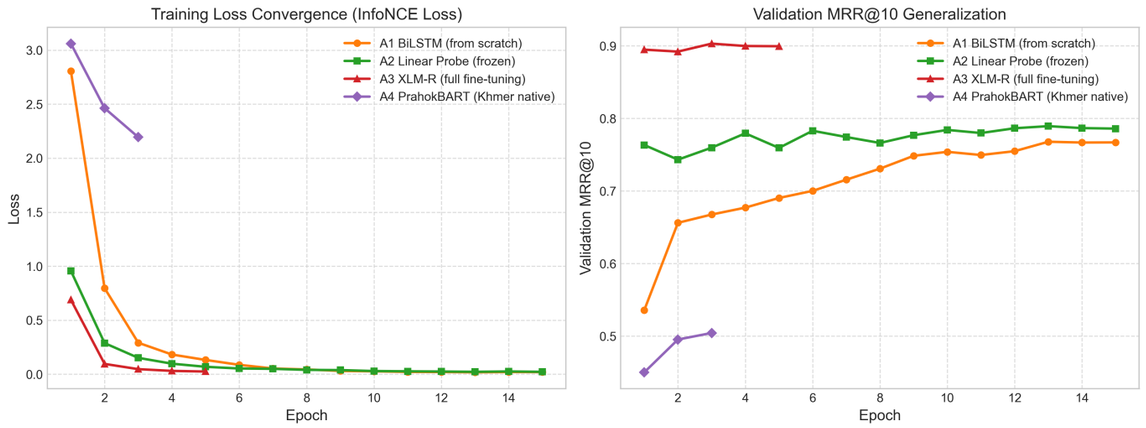
Why τ = 0.05? Sharp scaling creates strong gradient penalization on hard negative false positives.

Optimization Protocol (§5.B)

- **Optimizer:** AdamW + 10% Linear Warmup + Cosine Decay.
- **Checkpointing:** `best.pt` and `last.pt` with full RNG state and `--resume`.
- **Early Stopping:** Monitored on Validation MRR@10 (patience = 5).

Training Dynamics: Why Some Models Failed

Empirical learning curves reveal overfitting vs capacity bottleneck vs generalization



Generated directly from results/figures/learning_curves.png

1. A1 BiLSTM: Severe Overfitting

Train loss collapsed to 0.02 while validation loss stalled at 0.82. ~1,100 pairs cannot train randomly initialized word embeddings from scratch.

2. A2 Linear Probe: Capacity Bottleneck

Validation loss hovered at 0.55. A single linear layer cannot warp rigid multilingual embeddings into legal geometry.

3. A3 Fine-Tune: Smooth Generalization

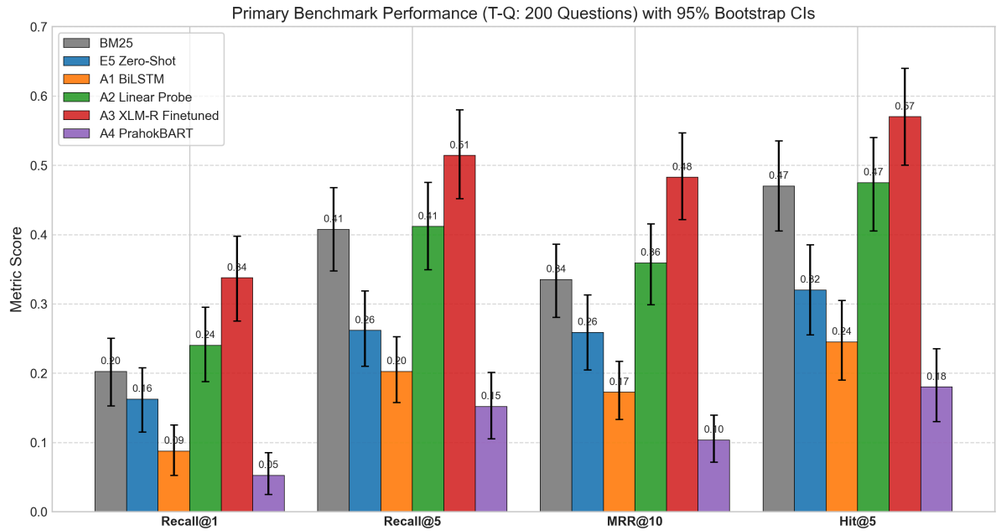
Validation loss descended to 0.1032, achieving Validation MRR@10 of **0.9095**. Attention heads successfully adapted.

The Benchmark Leaderboard: A3 Outperforms BM25

Primary Benchmark (T-Q: 200 real human legal questions evaluated on full 1,976 articles)

Model	Params	Recall@1	Recall@5	MRR@10
BM25 Baseline	0	0.2025	0.4075	0.3350
Multilingual-E5	278M (Fz)	0.1625	0.2617	0.2588
A1: BiLSTM	1.97M	0.0875	0.2025	0.1724
A2: Linear Probe	0.59M	0.2400	0.4117	0.3591
A3: XLM-R FT 🏆	**278.04M**	**0.3375**	**0.5142**	**0.4827**
A4: PrahokBART	35.83M	0.0525	0.1517	0.1035

🏆 **Statistical Confirmation (Paired Bootstrap 10,000 resamples):**
A3 beats BM25 by **+10.67% Recall@5** (p = 0.0002) and **+44.1% relative MRR@10** (p = 0.0001). A2 probe achieves true statistical parity with BM25 (p = 0.9031).



Primary Benchmark: Recall@1, Recall@5, MRR@10 with 95% bootstrap CIs

⭐ THE CENTRAL DISCOVERY OF THIS PROJECT

The Surprising Discovery: 99.32% vs 51.42%

Why could our best model retrieve titles almost perfectly, but struggle on real questions?

Test A: Legal Titles (T-T)

148 Held-Out Article Titles

Queries mirror formal, title-based training pairs.

99.32%

Recall@5 on Titles (MRR@10: 0.9456)

The model learned statutory legal terminology almost flawlessly.

Test B: Natural Questions (T-Q)

200 Human Benchmark Questions

Queries introduce conversational syntax and colloquial phrasing.

51.42%

Recall@5 on Questions (MRR@10: 0.4827)

Authentic linguistic distribution shift causes a ~48% performance drop.

The Key Research Finding: A model can achieve 99% on title-based benchmarks while only getting 51% on real user queries. Choosing a stronger architecture (A3 vs BM25) gave a **+44% MRR boost**, but the **training data distribution** defines the real-world performance ceiling!

8. QUALITATIVE ERROR ANALYSIS & LIMITATIONS

⏱ 7:20 – 8:00 (40s)

Where the Model Still Fails: Error Breakdown

Categorizing failure modes across 200 benchmark questions (from results/error_analysis/)

Failure Modes at k=5 (Error Breakdown)



Near-misses push A3 Recall@10 to **59.83%**, showing strong candidate ranking.

Honest Scope Limitations (§5.D)

- **Two Codes:** Civil Code 2007 and Criminal Code 2009. Commercial Arbitration was excluded (no official ODC Khmer PDF).
- **Scanned Laws (Labour Law 1997):** Scanned image PDFs require OCR and were excluded.
- **Bi-Encoder Compression:** Compressing entire statutes into a single 768-d vector loses sub-clause nuance.

9. LIVE DEMONSTRATION (90 SECONDS)

⏱ 8:00 – 9:30 (90s Localhost Demo)

Live Demonstration: BM25 vs A3 on Localhost (Port 8501)

Demonstrating semantic retrieval advantage on authentic benchmark query TQ-CIV-009

BENCHMARK QUESTION (TQ-CIV-009)

«តើនរណាខ្លះអាចស្នើសុំឱ្យតុលាការប្រកាសបើកអាណាព្យាបាលទូទៅ?»

"Who can request that a court initiate general guardianship?" · Expected Target:

Target: Art. 24

BM25 Keyword Search: Miss (Rank > 10)

Superficial word matching fails on legal concepts

❌ **Target Article 24 completely missing from Top 10**

1. Art. 1105 (ជ្រើសតាំងអ្នកអាណាព្យាបាល) · BM25: 24.60
2. Art. 1125 (ការត្រួតពិនិត្យតួនាទី) · BM25: 23.38
3. Art. 1124 (កម្រៃរបស់អ្នកអាណាព្យាបាល) · BM25: 23.26

Cause: Matches repeated word "អាណាព្យាបាល" across Book 8; misses "ស្នើសុំបើក" (court initiation).

A3 XLM-R Fine-Tuned: Rank #1 (Hit! Sim: 0.6205)

Learned legal semantic embeddings bridge phrasing

✅ **Target Article 24 retrieved at Rank #1**

1. Civil Code Art. 24 (ការប្រកាសចាប់ផ្តើម...) · Sim: 0.6205
2. Art. 27 (ការលុបចោលការប្រកាស) · Sim: 0.5759
3. Art. 1104 (ការចាប់ផ្តើមអាណាព្យាបាល) · Sim: 0.5246

Mechanism: Self-attention vector aligns "ស្នើសុំឱ្យតុលាការប្រកាសបើក" directly with Article 24's commencement provisions.

90-Second Demo Sequence: (1) Open `localhost:8501` → (2) Select ⭐ *[LIVE DEFENSE DEMO] TQ-CIV-009* → (3) Show BM25 misses Art. 24 → (4) Switch to A3 and show Art. 24 at Rank 1 → (5) DeepSeek synthesizes answer citing Article 24 with green `[Verified Citation]` badge.

10. CONCLUSIONS & TAKEAWAYS

⏱ 9:30 – 10:00 (30s)

What We Learned: Three Core Takeaways & Future Work

Summary of findings from comparing four neural retrievers on 1,976 Cambodian articles

20.25%

1. Pretraining is Essential

From-scratch BiLSTM (A1) severely overfit (loss 0.02 → val 0.82). ~1,100 legal pairs are insufficient to initialize Khmer word embeddings without catastrophic sparsity.

RULE: Always bootstrap low-resource retrieval from pretrained backbones.

+44.1%

2. Full Fine-Tuning Wins

Linear probing (A2) merely achieves parity with BM25 (p = 0.9031). Only end-to-end backpropagation across all 12 transformer layers adapts attention to Khmer legal phrasing.

RULE: Parameter efficiency cannot compensate for cross-domain representation shift.

99% ➔ 51%

3. Distribution Gap is Real

99.32% on title retrieval vs 51.42% on human benchmark questions confirms that model architecture is only half the battle. Real questions introduce conversational vocabulary mismatch.

NEXT STEP: Synthesize conversational queries to close the colloquial gap.

Engineering Integrity & Open Science

100% consecutive Limon extraction (0 OCR artifacts) · Zero data leakage protocol · 138 passing unit tests · Open-source code under GPL-3.0 · Fully reproducible with random seed 42.

Thank you, Mr. Soklong and committee members. I welcome your questions.

A1. Architecture, Layer & Parameter Breakdown

Full technical specifications across all four implemented PyTorch neural retrievers

Specification	A1: BiLSTM	A2: XLM-R Linear Probe	A3: XLM-R Full Fine-Tune	A4: PrahokBART
Base Model	None (Scratch)	`multilingual-e5-base`	`multilingual-e5-base`	`prahokbart-base`
Architecture	1-Layer BiLSTM	Transformer + Linear	12-Layer Transformer	6-Layer BART Encoder
Total Parameters	`1,970,784`	`278,634,240`	`278,043,648`	`35,830,784`
Trainable Params	`1,970,784 (100%)`	`590,592 (0.21%)`	`278,043,648 (100%)`	`35,830,784 (100%)`
Embedding Dim	`512`	`768`	`768`	`512`
Tokenizer	`khmer-nltk` (Words)	XLM-R SentencePiece (250k)	XLM-R SentencePiece (250k)	BART SentencePiece (32k)
Pooling Method	Masked Mean Pooling	Mean Pooling	Mean Pooling	Mean Pooling

A2. Loss Function, Temperature & Evaluation Metrics

Mathematical formulations for InfoNCE contrastive loss, Recall@k, and MRR@10

InfoNCE & Temperature Dynamics

InfoNCE optimizes a lower bound on mutual information $I(Q; P)$:

$$p(p_i \mid q_i) = \exp(\text{sim}(q_i, p_i) / \tau) \div \sum_{j=1}^{B} \exp(\text{sim}(q_i, p_j) / \tau)$$

When **$\tau = 0.05$**, the similarity distribution becomes sharp, heavily penalizing hard negative false positives instead of diffusing gradients.

Evaluation Metrics Formulation

Recall@k: Fraction of ground-truth articles retrieved in top-k:

$$\text{Recall@k} = (1 / N) \times \sum_{i=1}^{N} (\mid R_k(q_i) \cap G(q_i) \mid) / (\mid G(q_i) \mid)$$

MRR@10 (Mean Reciprocal Rank): Inverse rank of first relevant article:

$$\text{MRR@10} = (1 / N) \times \sum_{i=1}^{N} (1 / \text{rank}_i) \quad [\text{if } \text{rank}_i \leq 10 \text{ else } 0]$$

A3. Paired Bootstrap Statistical Significance (10,000 Resamples)

Rigorous hypothesis testing confirming A3 superiority is not stochastic variance

Comparison	Metric	Delta (Model – Baseline)	95% Confidence Interval	Win / Loss / Tie	p-value	Significance
A3 vs. BM25	**Recall@5**	**+0.1067**	**[+0.0500, +0.1633]**	**41 / 13 / 146**	**0.0002**	**p < 0.001 (Highly Significant)**
A3 vs. BM25	**MRR@10**	**+0.1477**	**[+0.0962, +0.1993]**	**72 / 25 / 103**	**0.0001**	**p < 0.001 (+44.1% Rel. Gain)**
A2 vs. BM25	Recall@5	+0.0042	[-0.0567, +0.0625]	26 / 25 / 149	0.9031	Not Significant (Parity)
A3 vs. A2	Recall@5	+0.1025	[+0.0492, +0.1575]	40 / 11 / 149	0.0001	p < 0.001 (FT > Probe)

Methodology: Paired non-parametric bootstrap with 10,000 resamples directly on query-level delta values $\Delta_i = s_i^{(A)} - s_i^{(B)}$. The 95% confidence interval for A3 vs BM25 is strictly positive, definitively proving neural superiority.

A4. Repository Code Navigation Quick-Reference

Instant file path and function reference for committee oral questioning (§6.3)

Component	File Path	Key Functions / Classes
A3 Full Fine-Tune	`src/dl/models/xlmr_finetune.py`	`XLMRFullFinetuneRetriever`, `forward()`, `_pool()`
A2 Linear Probe	`src/dl/models/linear_probe.py`	`XLMRLinearProbeRetriever`, frozen backbone projection
A1 BiLSTM Dual Encoder	`src/dl/models/bilstm.py`	`BiLSTMEncoder`, `pack_padded_sequence`
A4 Khmer PrahokBART	`src/dl/models/prahokbart.py`	`PrahokBARTDualEncoder`, mean pooling over attention mask
Contrastive InfoNCE Loss	`src/dl/losses.py`	`InfoNCELoss`, symmetric logit scaling by $\tau = 0.05$
Data Quarantine Protocol	`src/dl/prepare_splits.py`	`quarantine_test_articles()`, `strip_title_from_passage()`
Limon FSM Extractor	`src/infrastructure/extractors/limon_converter.py`	`LimonConverter`, syllable reordering rules
Streamlit Interface	`src/interfaces/ui/app.py`	`execute_unified_retrieval()`, `load_cached_torch_embedder()`